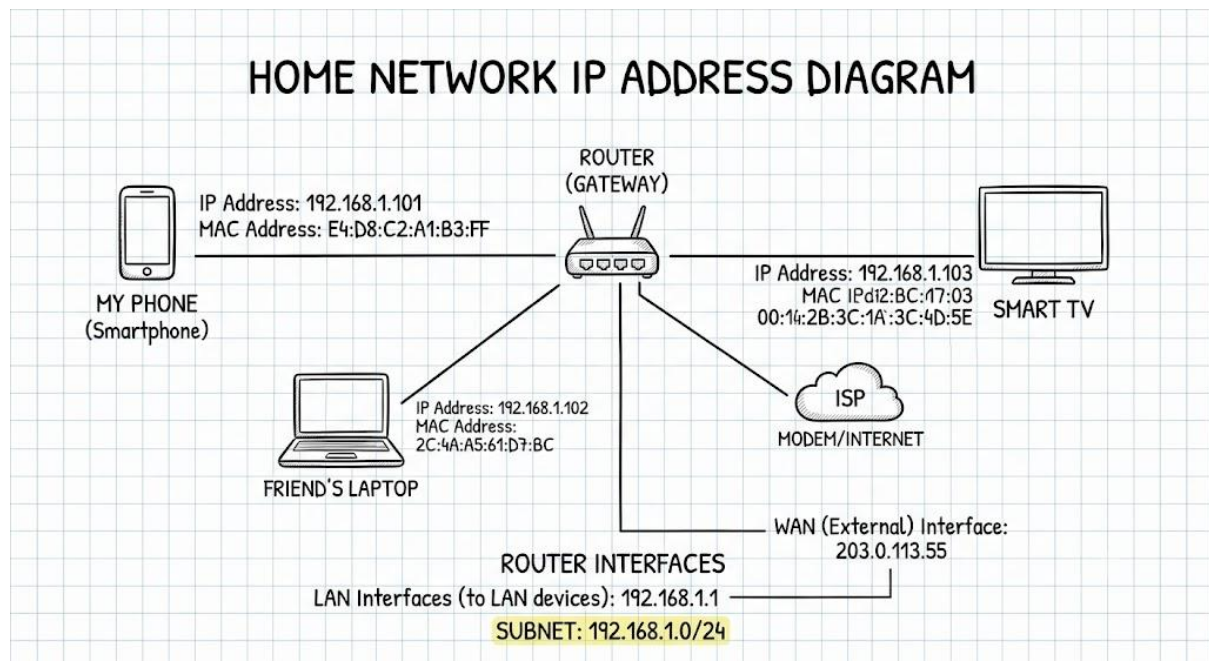


TASK-1



TASK-2

Your simulation is well structured. Here's a cleaner, assignment-ready version with a simple routing summary table at the end.

IP Packet Journey Across Two Routers

1. Network Setup & Topology

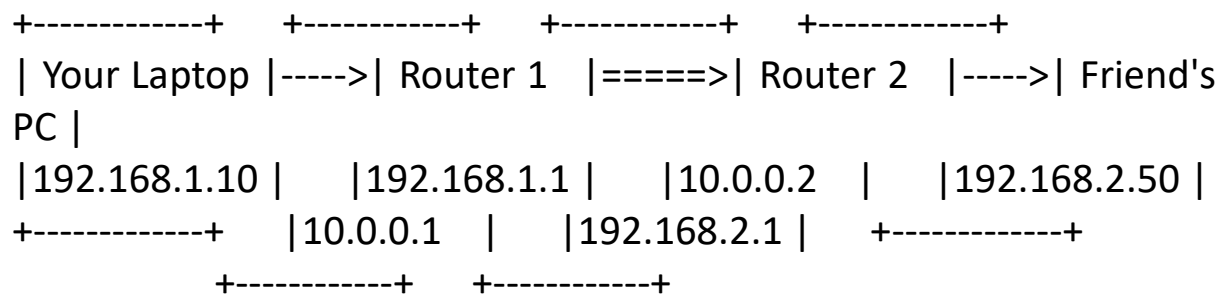
- **Your Laptop (Source):** 192.168.1.10/24
- **Router 1 (R1)**
 - LAN: 192.168.1.1
 - WAN: 10.0.0.1/30
- **Router 2 (R2)**
 - WAN: 10.0.0.2/30
 - LAN: 192.168.2.1

Friend's Laptop (Destination): 192.168.2.50/24

LAN

WAN

LAN



Step 1: Source Laptop

1. The laptop wants to send data to **192.168.2.50**.
2. It compares the destination with its subnet (**192.168.1.0/24**).
3. Since the destination is in a different subnet, it sends the packet to the **default gateway (192.168.1.1)**.
4. The packet is placed inside an Ethernet frame and sent to **Router 1**.

Step 2: Router 1 (R1)

1. R1 removes the Ethernet header and checks the destination IP.
2. It looks up its routing table.
3. It finds that network **192.168.2.0/24** is reachable through **10.0.0.2**.
4. R1 decreases the **TTL** by 1.
5. The packet is encapsulated in a new frame and forwarded through the WAN to **Router 2**.

Step 3: Router 2 (R2)

1. Router 2 receives the packet on its WAN interface.
2. It checks its routing table.
3. The destination network **192.168.2.0/24** is directly connected.
4. R2 performs an **ARP** request (if necessary) to obtain the MAC address of **192.168.2.50**.
5. The TTL is decreased by 1.
6. The packet is sent onto the local LAN.

Step 4: Destination Laptop

1. The friend's laptop receives the Ethernet frame.

2. It verifies that the destination IP is **192.168.2.50**.
3. The Ethernet and IP headers are removed.
4. The data is delivered to the correct application (such as Ping, File Sharing, or a Web Browser).

Routing Summary

Device	Source IP	Destination IP	Next Hop	Outgoing Interface
Your Laptop	192.168.1.10	192.168.2.50	192.168.1.1	eth0
Router 1	192.168.1.10	192.168.2.50	10.0.0.2	eth1
Router 2	192.168.1.10	192.168.2.50	Directly Connected	eth0
Friend's Laptop	192.168.1.10	192.168.2.50	Packet Received	NIC

Conclusion

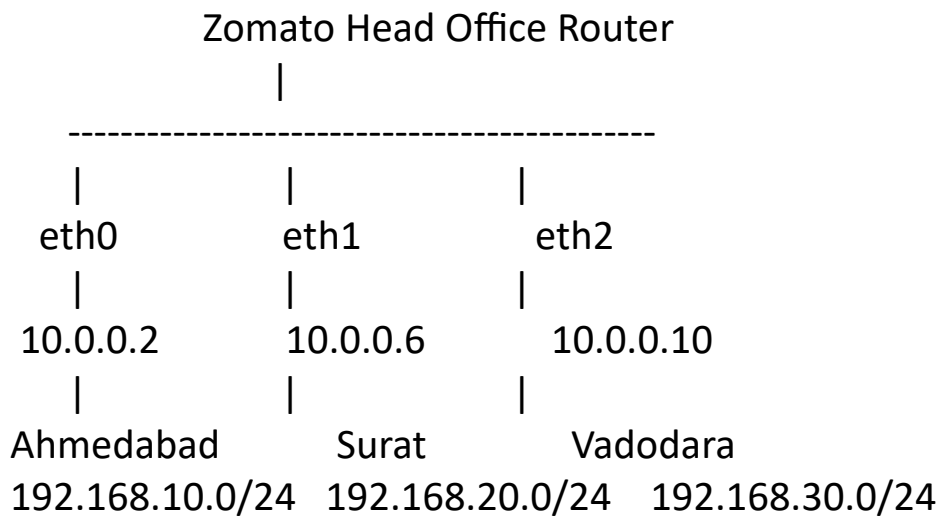
The IP packet travels from the source laptop to **Router 1**, then to **Router 2**, and finally reaches the friend's laptop. At each router, the routing table is checked, the **TTL** is reduced by one, and a new Layer 2 frame is created for the next hop until the packet reaches its destination.

TASK-3

Routing Table for a Zomato-Style Delivery Network

Branch (City)	Network Address	Subnet Mask	Next Hop	Interface
Ahmedabad Branch	192.168.10.0	255.255.255.0	10.0.0.2	eth0
Surat Branch	192.168.20.0	255.255.255.0	10.0.0.6	eth1
Vadodara Branch	192.168.30.0	255.255.255.0	10.0.0.10	eth2

Network Diagram



Explanation

- **Ahmedabad Branch:** Uses subnet **192.168.10.0/24** and is reached through **10.0.0.2** via **eth0**.
- **Surat Branch:** Uses subnet **192.168.20.0/24** and is reached through **10.0.0.6** via **eth1**.
- **Vadodara Branch:** Uses subnet **192.168.30.0/24** and is reached through **10.0.0.10** via **eth2**.

This routing table allows the head office router to forward delivery and order data to the correct city branch efficiently.

TASK -4

Routing Type	Pros (3)	Cons (3)	Real-World Example
Static Routing	1. Simple to configure.2. Low CPU and bandwidth usage.3. More secure (no routing updates).	1. Manual configuration required.2. Not suitable for large networks.3. Does not automatically adapt to network failures.	Small office or a small restaurant using Zomato POS with a simple network.

Routing Type	Pros (3)	Cons (3)	Real-World Example
Dynamic Routing (RIP/OSPF)	1. Automatically updates routes. 2. Scales well for large networks. 3. Automatically finds alternate paths if a link fails.	1. More complex to configure. 2. Uses CPU and bandwidth for routing updates. 3. Requires more memory and processing power.	Zomato, Google Maps, or YouTube data centers use dynamic routing to keep services fast and available even if a network path fails.

TASK-5

Sample OSPF Configuration Between Two Routers (Cisco IOS)

Router 1 (R1)

```
enable
configure terminal
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
end
write memory
```

Router 2 (R2)

```
enable
configure terminal
router ospf 1
network 192.168.2.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
end
write memory
```

Explanation of Each Configuration Line

Configuration Line	Description
enable	Enters privileged EXEC mode.
configure terminal	Enters global configuration mode.

Configuration Line	Description
router ospf 1	Starts the OSPF routing process with Process ID 1 .
network 192.168.1.0 0.0.0.255 area 0	Enables OSPF for the 192.168.1.0/24 network and places it in Area 0 .
network 192.168.2.0 0.0.0.255 area 0	Enables OSPF for the 192.168.2.0/24 network on Router 2.
network 10.0.0.0 0.0.0.3 area 0	Enables OSPF on the link connecting the two routers (10.0.0.0/30).
end	Exits configuration mode.
write memory	Saves the configuration so it remains after a restart.

In My Own Words

- enable gives administrator access to the router.
- configure terminal allows me to change the router's settings.
- router ospf 1 turns on the OSPF routing protocol.
- The network commands tell OSPF which networks should be advertised to other routers.
- area 0 places the networks in the OSPF backbone area.
- end exits configuration mode.
- write memory saves the configuration permanently.